ModelIt!

Teacher's Guide

Why Can't You See Your Own Cells?

Exploring the Microscopic Building Blocks of Life

Quick Reference

Grade Level	Grade 6
NGSS Standard	MS-LS1-1, MS-LS1-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-LS1-1, MS-LS1-2

Conduct an investigation to provide evidence that living things are made of cells; either one cell or many different numbers and types of cells. Develop and use a model to describe the function of a cell as a whole and ways the parts of cells contribute to the function.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model showing how cell size, nutrients, oxygen, and waste interact to determine cell health.
Disciplinary Core Idea	**LS1.A Structure and Function** — All living things are made up of cells, which is the smallest unit that can be said to be alive. Cells have specific structures that perform specific functions.
Crosscutting Concept	**Scale, Proportion, and Quantity** — Students explore why cells must remain at microscopic scales to function — connecting size to the efficiency of material exchange.

CAST Testing Alignment

MULTIPLE CHOICE:

A scientist observes that cells in a low-oxygen environment stop growing and begin to die. Which best explains this observation based on cell structure and function?

CONSTRUCTED RESPONSE:

Using what you know about cells, explain why a single cell cannot grow to be as large as a golf ball. Use the concepts of surface area, volume, and nutrient exchange in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

Cells are the fundamental units of life — every living organism on Earth is made of one or more cells. Understanding cell structure and function is the foundation of all biology and medicine.

Cell Theory

All living things are composed of cells. Cells are the basic unit of structure and function in living things. All cells come from pre-existing cells. This theory, developed in the 1830s-1850s by Schleiden, Schwann, and Virchow, is one of the foundational principles of modern biology.

Why Cells Are Small

Cells must exchange materials (nutrients in, waste out) across their cell membrane. As a cell grows, its volume increases faster than its surface area (cube vs. square relationship). Eventually, the membrane can't supply the interior fast enough. This surface-area-to-volume ratio is the key constraint on cell size.

Cell Organelles

Key organelles include the nucleus (DNA storage and control center), mitochondria (energy production), ribosomes (protein construction), endoplasmic reticulum (protein and lipid processing), and cell membrane (boundary and gatekeeper). Each plays a critical role in keeping the cell alive.

Cells in the Human Body

The human body contains approximately 37.2 trillion cells of about 200 different types. Red blood cells carry oxygen, white blood cells fight infection, neurons transmit signals, and muscle cells enable movement. Despite their diversity, all share the same basic cellular machinery.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify cell size, nutrient supply, oxygen level, and waste buildup as key components affecting cell function.

E **Establish Relationships**
Students determine that nutrient supply positively affects cell function while waste buildup negatively affects it, and that oxygen is essential for energy production.

V **Visualize & Model**
Students build a model showing how inputs (nutrients, oxygen) and outputs (waste) determine whether a cell thrives, shrinks, or dies.

E **Evaluate Outcomes**
Students run starvation and oxygen-crisis scenarios to observe how removing key inputs affects the entire system.

R **Revise & Extend**
Students add cell division rate or temperature to explore more complex interactions.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Did you know your body has more cells than there are stars in the Milky Way?"*

Watch For

Show a quick video or image of cells under a microscope to build wonder.

Slide 2: Learning Objectives

3 min

Say: *"Today we're going to explore the invisible world inside YOU."*

Watch For

Have students read objectives aloud. Pre-teach 'organelle' and 'cell membrane.'

Slide 3: Big Question

3 min

Say: *"Think about this: right now, trillions of cells are working inside you and you can't see ANY of them."*

Watch For

Have students turn and talk: What's the smallest thing you've ever seen?

Slide 4: LEVER Framework

2 min

Say: *"We'll build a model to understand why cells MUST be tiny."*

Watch For

Briefly preview each LEVER step. Connect to prior knowledge of models.

Slide 5: Activity 1: Components

8 min

Say: *"What does a cell need to survive? What threatens it?"*

Watch For

Guide sorting. Discuss why nutrient supply and oxygen come from OUTSIDE the cell.

Slide 6: Activity 2: Connections

8 min

Say: *"More nutrients help cells grow — but what happens when waste piles up?"*

Watch For

Help students identify positive and negative relationships. Draw arrows.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's starve a cell and see what our model predicts!"*

Watch For

Have students predict first, then run simulations. Compare healthy vs. starved cells.

Slide 8: Discoveries

5 min

Say: *"So WHY can't cells be huge? What did our model reveal?"*

Watch For

Lead discussion connecting surface-area-to-volume ratio to their model results.

Slide 9: STEM Challenge

12 min

Say: *"Now you're city planners — but your city IS a cell!"*

Watch For

Distribute materials. Groups design Cell City showing resource flow and waste removal.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Nutrient Supply, Oxygen Level
- **INTERNAL:** Cell Size, Waste Buildup

Relationships

- **Nutrient Supply to Cell Size:** POSITIVE (+) — More nutrients allow the cell to grow and maintain its structures. Without nutrients, cells shrink and stop dividing.
- **Oxygen Level to Waste Buildup:** NEGATIVE (-) — Higher oxygen levels allow cellular respiration to work efficiently, breaking down waste. Low oxygen causes waste to accumulate.
- **Waste Buildup to Cell Size:** NEGATIVE (-) — As waste accumulates, it becomes toxic and damages cell structures, causing cells to shrink and eventually die.

Key Discoveries

- Cells must stay small because nutrients and oxygen can only diffuse a short distance
- When nutrient supply drops, cells shrink and may stop dividing entirely
- Waste buildup is toxic — cells that can't remove waste lose function and die
- The surface-area-to-volume ratio explains why cells can't grow infinitely large

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Cells are just blobs of jelly"	Cells are highly organized structures with specialized parts (organelles) that work together like a factory. Each organelle has a specific job critical to the cell's survival.
"All cells look the same"	Human bodies have about 200 different cell types, each with a unique shape matched to its function. Red blood cells are disc-shaped for gas exchange, neurons are long for signal transmission, and muscle cells are fiber-shaped for contraction.
"Cells are not alive"	Each cell independently carries out ALL life functions: obtaining energy, removing waste, growing, reproducing, and responding to its environment. A single-celled organism like an amoeba IS a complete living thing.